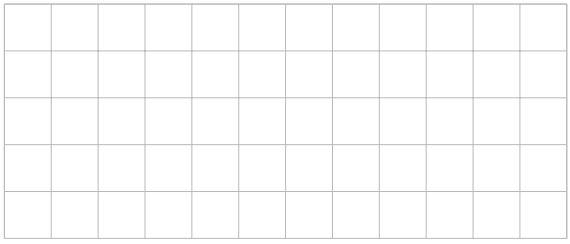
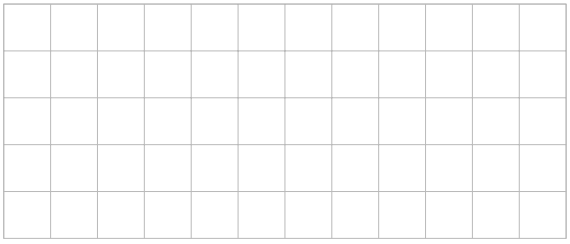
Problem 3: A shared side is where two tiles meet. Use ten tiles to make shapes with 9, 10, 11, 12, and 13 shared sides. Find a rule that gives the boundary length from the number of tiles and the number of shared sides.



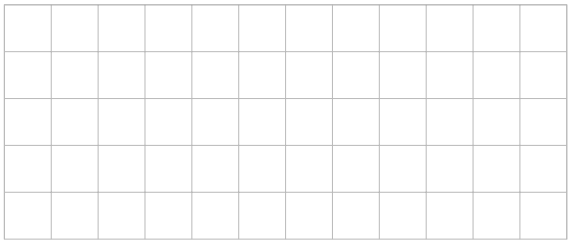
9 shared sides

Boundary: _____



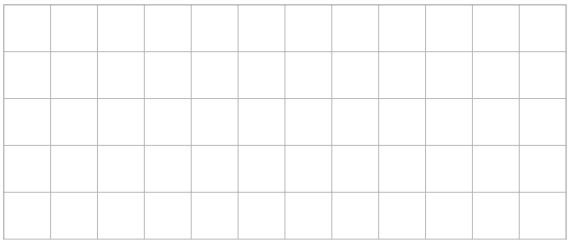
10 shared sides

Boundary: _____



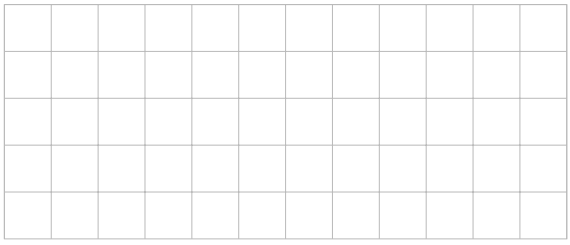
11 shared sides

Boundary: _____



12 shared sides

Boundary: _____

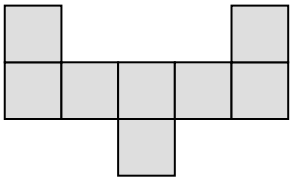


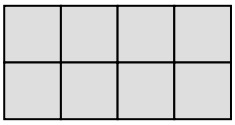
13 shared sides

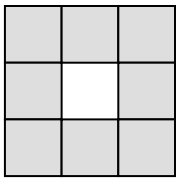
Boundary: _____

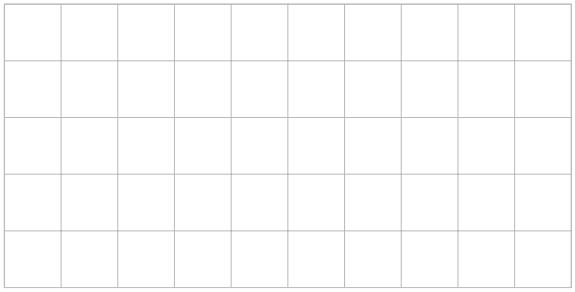
Problem 5: Which of these eight-tile shapes have the longest possible boundary? Explain why each chosen shape reaches that length. Draw two more such shapes, with no straight row or column of four tiles.

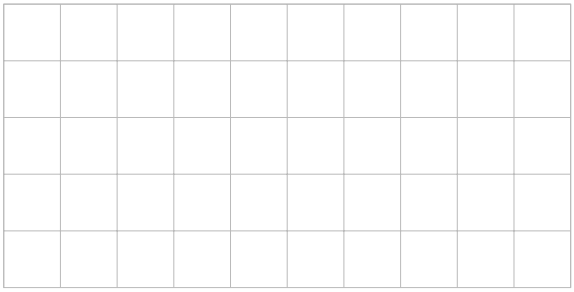




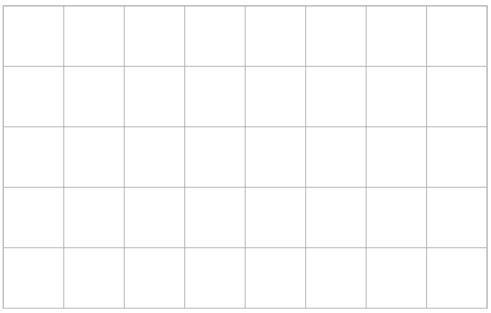






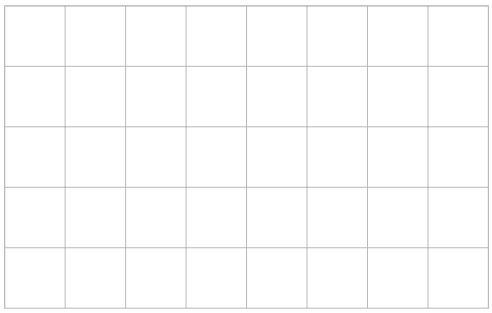


Problem 6: Add one tile to a shape so that its boundary changes by each amount shown. For each amount, use as few starting tiles as possible. Explain why fewer tiles cannot work.



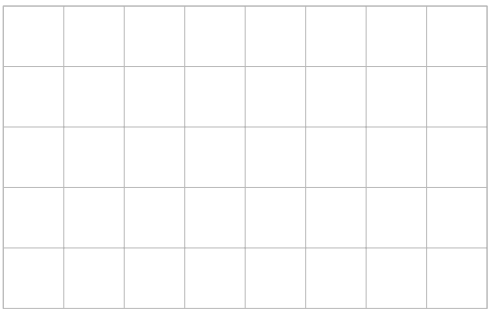
Boundary change: $+2$

Starting tiles: _____



Boundary change: 0

Starting tiles: _____



Boundary change: -2

Starting tiles: _____



Boundary change: -4

Starting tiles: _____